Advance Python Assignment

Topic-5: Exception Handling

Q1. Explain exception handling in Python using try, except, and finally.

Ans.
Exception handling in Python is used to manage errors that occur during the execution of a program. It prevents the program from stopping suddenly and allows it to run smoothly. The try block is used to write code that may cause an error, while the except block is used to handle that error. The finally block is executed whether an error occurs or not. This helps in proper execution and cleanup of resources. Exception handling makes programs more stable and user-friendly.

Example:
```
try:
   x = int(input("Enter number: "))
   print(100 / x)
except ZeroDivisionError:
   print("Error: division by zero")
finally:
   print("Program ended")
```

Q2. Explain handling multiple and user-defined exceptions in Python.

Ans.
In Python, multiple exceptions can be handled using more than one except block. This allows the program to react differently to different types of errors. User-defined exceptions are created by the programmer to handle special situations. They are created using classes and help in customizing error handling. This improves control over the program and makes error messages more meaningful.

Example:
```
try:
   value = int(input("Enter value: "))
   print(50 / value)
except ZeroDivisionError:
   print("Cannot divide by zero")
```

```
except ValueError:
    print("Invalid input")

class CustomError(Exception):
    pass

try:
    raise CustomError("This is a custom error")
except CustomError as e:
    print(e)
```